

Practice Quiz: Cognition

Part A: Multiple Choice

1. In Active Inference, a "belief" is best understood as: A) A sentence in a language of thought B) A posterior probability distribution over hidden states, physically realized in internal states C) A subjective feeling of confidence about a proposition D) A behavioral disposition to act in a certain way
2. Precision in Active Inference plays the functional role of: A) Memory consolidation B) Attention — determining which prediction errors are amplified C) Emotion — generating affective valence D) Motor planning — selecting the best action sequence
3. The dark room problem challenges Active Inference by asking: A) How agents can perceive in dark environments B) Why agents don't seek out maximally predictable environments to minimize surprise C) Whether light is necessary for generative models to function D) How sleep relates to free energy minimization
4. The resolution of the dark room problem relies on: A) The claim that agents also maximize entropy B) Prior preferences (C vector) that make a dark room surprising relative to the agent's model C) The physical impossibility of finding a completely dark room D) The distinction between conscious and unconscious agents
5. According to the Active Inference view, aberrant precision may explain: A) Viral infections B) Conditions like anxiety (overweighting threat) and autism (overweighting sensory prediction errors) C) Normal aging D) Evolutionary adaptation
6. The "hard problem of consciousness" (Chalmers) asks: A) How the brain generates neural activity B) Why physical processes give rise to subjective experience C) Whether AI can be intelligent D) How to measure brain activity using fMRI
7. If beliefs are physical states of a generative model, then the extended mind hypothesis implies: A) External tools can never be part of cognition B) Beliefs can, in principle, be

physically instantiated in artifacts outside the brain C) Consciousness must extend to all physical objects D) Only digital devices can extend the mind

Part B: Short Answer

1. Explain how Active Inference accounts for the difference between habitual behavior and deliberate reasoning. What role does precision play?
2. A patient with blindsight denies seeing a stimulus but can "guess" its location above chance. Analyze this case from the Active Inference perspective: does the patient have a "belief" about the stimulus? What does this reveal about the relationship between inference and consciousness?
3. Construct a philosophical argument for or against the claim: "If the self is a belief (an inference), it is no less real than if it were a substance."